

GLASGOW COLLEGE UESTC

Exam paper

Application and Design of Digital Logic (UESTC 2001)

Date: (remember to complete when info available from Ruoli)

Time: (remember to complete when info available from Ruoli)

Attempt all PARTS. Total **100** marks

Use one answer sheet for each of the questions in this exam.

Show all work on the answer sheet.

For Multiple Choice Questions, use the dedicated answer sheet provided.

Make sure that your University of Glasgow and UESTC Student Identification Numbers are on all answer sheets.

An electronic calculator may be used provided that it does not allow text storage or display, or graphical display.

All graphs should be clearly labelled and sufficiently large so that all elements are easy to read.

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

Q1 Multiple choice questions (Only one choice is correct) [24]

- (a) For unsigned decimal number $(137)_{10}$, its 8-bit gray code is (). [2]
A. 11001101 B. 11110001 C. 10001001 D. 01110001 E. 01001101
- (b) In the following statements, the wrong one is (). [2]
A. $AB + A'C + BC = AB + A'C$
B. $A'C + BC + AB' + D = (BC)' + A'B + AC + D$
C. $(AB + CD)' = (A' + B')(C' + D')$
D. If $A + B = A + C$ and $AB = AC$, then $B = C$
E. If $AB = 0$ and $A + B = 1$, then $(A + C)(A' + B)(B + C) = BC$

The following three questions refer to the given $F = \sum_{ABC} (3,5,7)$.

- (c) The max-term list expression is $F = \prod_{ABC} ()$. [2]
A. 0,1,3,5,7 B. 1,3,5,6,7 C. 0,2,4 D. 3,5,7 E. 0,1,2,4,6
- (d) The complement expression is $F' = \prod_{ABC} ()$. [2]
A. 0,1,3,5,7 B. 1,3,5,6,7 C. 0,2,4 D. 3,5,7 E. 0,1,2,4,6
- (e) The duality expression is $F^d = \sum_{ABC} ()$. [2]
A. 0,1,3,5,7 B. 3,5,7 C. 0,2,4 D. 1,3,5,6,7 E. 0,1,2,4,6

The following two questions refer to a given 8-bit Full Adder hardware module, which input signals $A = (1101\ 1101)_2$, $B = (1011\ 0111)_2$, Carryin = $(1)_2$.

- (f) The output SUM of this Full Adder module should be (). [2]
A. 1001 0101 B. 0001 0101 C. 0001 0100 D. 1001 0100 E. 1001 0001
- (g) The output Carryout-flag and Overflow-flag of this Full Adder module should be (). [2]
A. 0, 0 B. 1, 1 C. 0, 1 D. 1, 0
E. depend on A&B are signed numbers or unsigned numbers

The following two questions refer to a given 1024*16bits RAM, which is used to build a combinational circuit.

- (h) The bit number of input signal of this combinational circuit at most is (). [2]
A. 8 B. 10 C. 16 D. 512 E. 1024
- (i) The bit number of output signal of this combinational circuit at most is (). [2]
A. 8 B. 10 C. 16 D. 512 E. 1024
- (j) N logic variables ($x_0, x_1, x_2, \dots, x_{N-1}$) participate in XOR operation. The result should be () if there are only 21 variables are equal to '1'. [2]
A. 0 B. 1 C. 21 D. N-1 E. N
- (k) For a 8-3 priority encoder, when the inputs are 0111 1000, its output is most likely to be (). Suppose all inputs are active high, and the rightmost MSB (Most Significant Bit) of input (encoded as '111') has the highest priority. [2]
A. 000 B. 001 C. 011 D. 111 E. 110

Continued overleaf

- (l) Figure Q1-1 shows a waveform of D-type flip-flop. The CP, R_D, S_D, V_i, Q connect to the clock, reset, set, D and Q pin separately. It can be inferred that the S_D most like a () signal. [2]

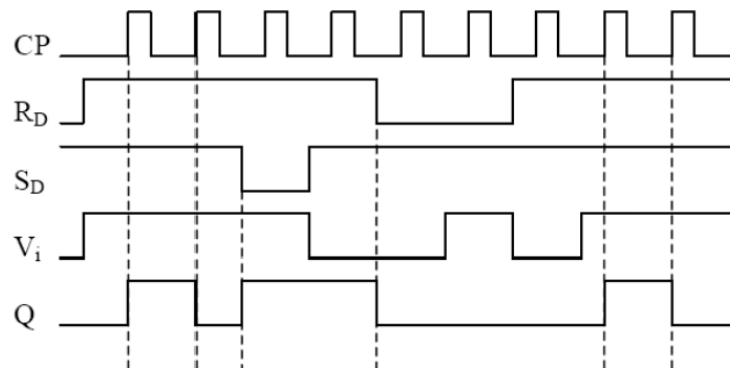


Figure Q1-1

- A. Synchronous, positive-edge-valid B. Asynchronous, positive-edge-valid
C. Synchronous, high-level-valid D. Asynchronous, negative-edge-valid
E. Asynchronous, low-level-valid

Q2 Given the logical function $F = \sum_{WXYZ}(8,9,12,13,14,15)$ [25]

- (a) Complete the Karnaugh-map in Figure Q1-1 for F. [6]

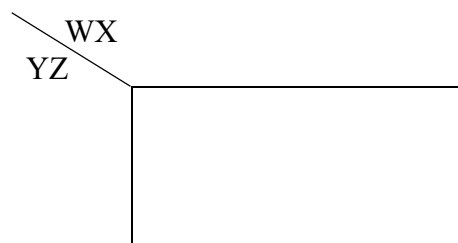


Figure Q2-1

- (b) Derive the Canonical Sum-of-Products (CSoP) expression for F. [6]

$$F(W,X,Y,Z) =$$

- (c) Derive the schematic diagram for F, with only one 74x138 (see Appendix on the last page) and one multi-input NAND gate. [9]
(d) Some people believe that the decoder (with several basic gates) can be used to implement any combinational circuit. Do you agree? Briefly describe your ideas. [4]

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- Q3** Analyze the sequential circuit in *Figure Q3-1*, which contains a 4-bit binary counter 74x163, a binary decoder 74x138 and an 8-input multiplexer 74x151. The function of this modules shown in Appendix on the last page. [25]

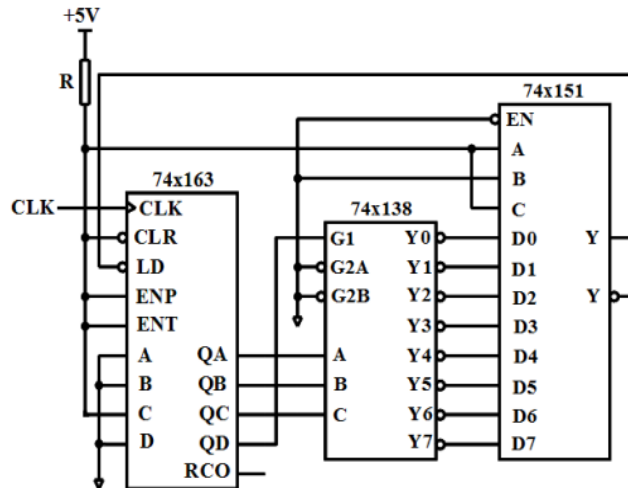


Figure Q3-1

- (a) Derive the logic equation for LD signal on 74x163, **with necessary notes**. [4]
 (b) Complete the State Transition/Output Table in *Figure Q3-2*. [8]

Current State QDQCQBQA	Next State QD*QC*QB*QA*	Output Y
0000		
0001		
0010		
0011		
0100		
0101		
0110		
0111		
1000		
1001		
1010		
1011		
1100		
1101		
1110		
1111		

Figure Q3-2

- (c) Derive the state transition graph (STG) based on part (b). Could you tell the function of this circuit? [9]
 (d) Is this circuit capable of self-correction? If so, briefly explain why. If the answer is NO, how can we change the design? [4]

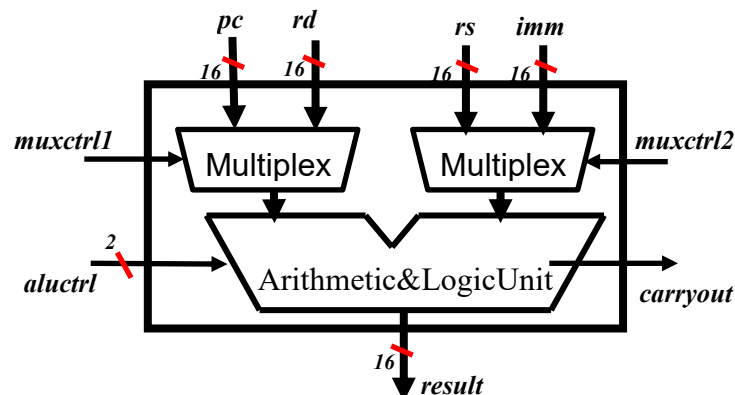
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- Q4** An instructions set of a simple RISC CPU is shown in *Figure.Q4-1*. *Rd* is a destination operator, and *Rs* is a source operator. ‘()’ indicates that *Rd* or *Rs* in parentheses is an address of a memory cell. *Imm* means an immediate data. [26]

Function description	Assembly instructions	Machine instructions			
		Opcode (4bit)	<i>Rd</i> (2bit)	<i>Rs</i> (2bit)	<i>Imm</i> (8bit)
$Rd \leftarrow Rs$	MOV <i>Rd</i> , <i>Rs</i>	000 0	xx	xx	Undefined
$Rd \leftarrow Imm$	MOV <i>Rd</i> , <i>Imm</i>	000 1	xx	Undefined	xxxxxxx
$Rd \leftarrow Rd + Rs$	ADD <i>Rd</i> , <i>Rs</i>	001 0	xx	xx	Undefined
$Rd \leftarrow Rd + Imm$	ADD <i>Rd</i> , <i>Imm</i>	001 1	xx	Undefined	xxxxxxx
$Rd \leftarrow Rd - Rs$	SUB <i>Rd</i> , <i>Rs</i>	010 0	xx	xx	Undefined
$Rd \leftarrow Rd - Imm$	SUB <i>Rd</i> , <i>Imm</i>	010 1	xx	Undefined	xxxxxxx
$Rd \leftarrow Rd \& Rs$	AND <i>Rd</i> , <i>Rs</i>	011 0	xx	xx	Undefined
$Rd \leftarrow Rd \& Imm$	AND <i>Rd</i> , <i>Imm</i>	011 1	xx	Undefined	xxxxxxx
$PC \leftarrow PC + Imm$ if carry flag is true	JC <i>Imm</i>	111 0	Undefined	Undefined	xxxxxxx
$PC \leftarrow PC + Imm$	JMP <i>Imm</i>	111 1	Undefined	Undefined	xxxxxxx

Figure.Q4-1

- (a) The ALUwMUX submodule in *Figure.Q4-2* can support all calculations in *Figure.Q4-1*, including PC update. Compose an HDL code for this ALUwMUX submodule, **with necessary notes**. [9]



Signal Name	Input or Output	Bit Width	Function Description
<i>pc</i>	input	16	first source data, come from PC register, for JMP/JC instruction
<i>rd</i>	input	16	first source data, come from Rd register
<i>rs</i>	input	16	second source data, come from Rs register
<i>imm</i>	input	16	second source data, come from instruction register
<i>result</i>	output	16	result data
<i>carryout</i>	output	1	=0 no carry, =1 carry, for JC instruction
<i>muxctrl1</i>	input	1	=0 select PC(for JMP/JC), =1 select rd
<i>muxctrl2</i>	input	1	=0 select rs, =1 select imm
<i>aluctrl</i>	input	2	=00 go through, =01 add, =10 sub, =11 add

Figure.Q4-2

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- (b) Compose a testbench in HDL for the ALUwMUX submodule in part (b), **with all necessary notes**. [9]
- (c) Sometimes we need use latches for CPU address bus for load/store instructions. The circuit diagram(a) and function table(b) of an SR latch shown in *Figure Q4-3*.

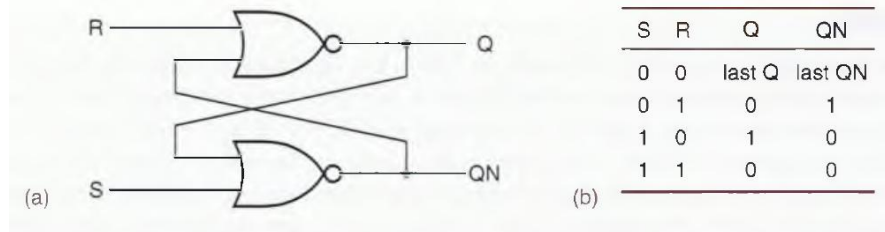


Figure. Q4-3

- (i) Complete the waveform in *Figure Q4-4*. Assume that the rise time and fall time of signal are zero and the propagation delay of the NOR gates is 10 ns. (Each division in *Figure Q4-4* is 10ns.) [6]

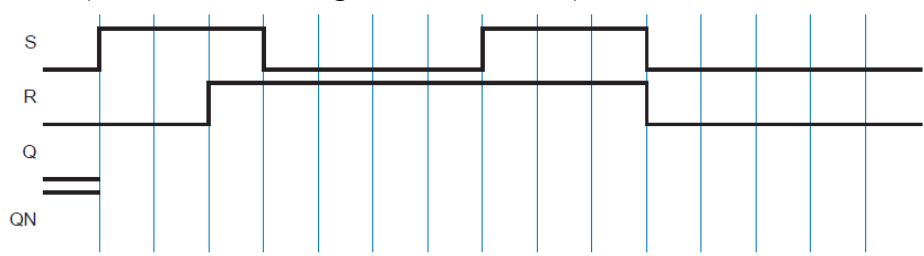
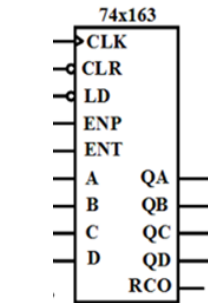


Figure Q4-4

- (ii) What can you observe from the waveforms in the part(b)? Briefly explain the phenomenon and describe the disadvantage of a SR latch. [2]

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Appendix

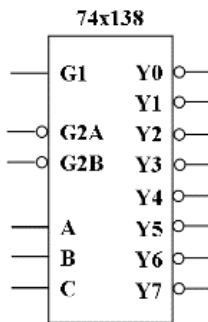


(a) Logic symbol

CLK	CLR	LD	ENP	ENT	QDQCQBQA	RCO
↑	0	X	X	X	0 0 0 0	0
↑	1	0	X	X	D C B A	0
↑	1	1	0	X	HOLD	0
↑	1	1	X	0	HOLD	0
↑	1	1	1	1	Count number+1 QD is MSB	1 When count number reaches '0000'

(b) Function table

Figure A1 Function of 74x163 (4-bit Binary Counter)

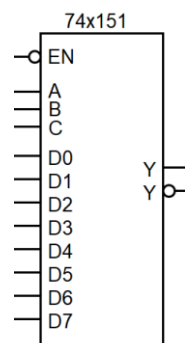


(a) Logic symbol

input				output							
Gx	GxA_L	GxB_L	C B A	Y7_L	Y6_L	Y5_L	Y4_L	Y3_L	Y2_L	Y1_L	Y0_L
0	X	X	X X X	1	1	1	1	1	1	1	1
X	1	X	X X X	1	1	1	1	1	1	1	1
X	X	1	X X X	1	1	1	1	1	1	1	1
1	0	0	0 0 0	1	1	1	1	1	1	1	0
1	0	0	0 0 1	1	1	1	1	1	1	0	1
1	0	0	0 1 0	1	1	1	1	1	0	1	1
1	0	0	0 1 1	1	1	1	1	0	1	1	1
1	0	0	1 0 0	1	1	1	0	1	1	1	1
1	0	0	1 0 1	1	1	0	1	1	1	1	1
1	0	0	1 1 0	1	0	1	1	1	1	1	1
1	0	0	1 1 1	0	1	1	1	1	1	1	1

(b) Truth table

Figure A2 Function of 74x138 (3-8 Decoder)



(a) Logic symbol

Input				Output
EN	C	B	A	Y
1	x	x	x	0
0	0	0	0	D0
0	0	0	1	D1
0	0	1	0	D2
0	0	1	1	D3
0	1	0	0	D4
0	1	0	1	D5
0	1	1	0	D6
0	1	1	1	D7

(b) Truth table

Figure A3 Function of 74x151 (8-1 Multiplexer)

End of question paper